

Dashboard dan Mindmap Interaktif Supply Demand

1. Dataset

- Perizinan

A	B	C	D	E	F	G	H	I	J	K	L	M	
id_izin	kode_perusahaan	jenis_badan	nama_perusahaan	jenis_perusa	kode_komod	komoditas	kode_kabkot	nama_kabko	kode_provin	nama_provinsi	kode_pulau	nama_pulau	
20780	14357	Perusahaan	Korr 3G TRUST	IUP	611	Pasir Kuarsa	1906	KAB. BELITUNG	19	KEPULAUAN BANGKA BELITUNG		1 SUMATERA	
21340	16620	Perusahaan	Pers AASAAK AFABA JAYA	IUP	512	Andesit	3514	KAB. PASURUAN	35	JAWA TIMUR		2 JAWA DAN B	
20572	16620	Perusahaan	Pers AASAAK AFABA JAYA	IUP	512	Andesit	3514	PASURUAN	35	JAWA TIMUR		2 JAWA DAN B	
22151	19988	Perusahaan	Pers ABAI DUA PUTRI	IUP	540	Kerikil Berpasir /	7601	KAB. PASANGKA	76	SULAWESI BARAT		4 SULAWESI	
23171	12289	Perusahaan	Pers ABAI MINERAL RESOL	IUP	533	Batu Gunung Qu	6302	KOTABARU	63	KALIMANTAN SELATAN		3 KALIMANTAN	
6238535841775	4779	Perusahaan	Pers Abadi Nikel Nusantara	IUP	212	Nikel	7206	KAB. MOROWAL	72	SULAWESI TENGAH		4 SULAWESI	
	11	16	Perusahaan	Pers ABAI OGAN CEMERLA	IUP	303	Batubara	1601	OGAN KOMERIN	16	SUMATERA		1 SUMATERA
	21585	18772	Perusahaan	Pers ABAI SILIKA PERSADA	IUP	611	Pasir Kuarsa	6207	KAB. SERUYAN	62	KALIMANTAN TENGAH		3 KALIMANTAN
	23142	21270	Perusahaan	Pers ABDARA BRIM TAMBAN	IUP	533	Batu Gunung Qu	7203	KAB. DONGGALA	72	SULAWESI TENGAH		4 SULAWESI
	13851	32	Perusahaan	Pers ABE JAYA PERKASA	IUP	303	Batubara	6205	KAB. BARITO UTI	62	KALIMANTAN TENGAH		3 KALIMANTAN
	18998	15800	Perusahaan	Korr ABHINAWA BERLIAN UTI	IUP	540	Kerikil Berpasir /	3508	KAB. LUMAJANG	35	JAWA TIMUR		2 JAWA DAN B
	20610	18613	Perusahaan	Pers ABHIPRAWA NAWASEN	IUP	611	Pasir Kuarsa	6101	KAB. SAMBAS	61	KALIMANTAN BARAT		3 KALIMANTAN
	22928	15766	Perusahaan	Korr ABIDIN JAYA UTAMA	IUP	540	Kerikil Berpasir /	3510	KAB. BANYUWANG	35	JAWA TIMUR		2 JAWA DAN B
	18686	36	Perusahaan	Pers ABISHA BUMI PERSADA	IUP	206	Emas	9207	KAB. TELUK WOI	92	PAPUA BARAT		5 PAPUA
	20990	13053	Perusahaan	Pers ACEH MINERAL PRIMA	IUP	429	Kalsit	1307	KAB. LIMA PULU	13	SUMATERA BARAT		1 SUMATERA
21641	11921	Perusahaan	Pers ACES SELARAS	IUP	533	Batu Gunung Qu	7271	KOTA PALU	72	SULAWESI TENGAH		4 SULAWESI	
23143	9564	Perusahaan	Pers ACK BARA ABAI	IUP	303	Batubara	6407	KAB. KUTAI BARU	64	KALIMANTAN TIMUR		3 KALIMANTAN	
22408	19723	Perusahaan	Pers ADANG PASER JAYA	IUP	512	Andesit	6401	KAB. PASER	64	KALIMANTAN TIMUR		3 KALIMANTAN	
23147	16939	Perusahaan	Pers ADARA JALA SAMUDRA	IUP	539	Pasir Pasang	1901	KAB. BANGKA	19	KEPULAUAN BANGKA BELITUNG		1 SUMATERA	
19810	53	Perusahaan	Pers ADARO INDONESIA	IUPK	303	Batubara	6300	TABALONG, BAL	63	KALIMANTAN SELATAN		3 KALIMANTAN	
10434	8259	Perusahaan	Pers ADE PUTRA TANRAJENI	IUP	303	Batubara	6402	KUTAI KARTANEI	64	KALIMANTAN TIMUR		3 KALIMANTAN	
20032	8261	Perusahaan	Pers ADE RESKY SEJAHTERA	IUP	505	Marmar	7316	KAB. ENREKANG	73	SULAWESI SELATAN		4 SULAWESI	
17537	9547	Perusahaan	Pers ADEI IN PENAIAM RORI	IUP	303	Batubara	6403	PENAIAM PASER	64	KALIMANTAN TIMUR		3 KALIMANTAN	
+ ≡ 🗺 Summary_1 Summary_2 Perizinan Data Dirilisasi Produksi Pemasaran_1 Pemasaran_2 Perusahaan_Cadangan Sumber Daya dan Cadangan 🔍 < >													

- Data Hilirisasi

	A	B	C	D	E	F	G	H	I	J	K	L
1	nama perusahaan											
2	PT Ocean Sky Metal Industry											
3	PT Metal Smetindo Selaras											
4	PT Halmahera Jaya Feronikel											
5	PT Virtus Dragon Nickel Industry											
6	PT Sunmy Metal Industry											
7	PT Kao Rahai Smelters											
8	PT Yashi Indonesia Investment											
9	PT Angel Nickel Industry											
10	PT Huadi Nickel - Alloy Indonesia											
11	PT Huake Nickel Indonesia											
12	PT Indonesia Guang Ching Nickel And Stainless Steel Industry											
13	PT Megah Surya Pertiwi											
14	PT Huaneng Metal Industry											
15	PT Indonesia Rapiu Nickel And Chrome Alloy											
16	PT Indonesia Toingshan Stainless Steel											
17	PT Indonesia Toingshan Stainless Steel											
18	PT Lipe Metal Industry											
19	PT Debonair Nickel Indonesia											
20	PT Infel Metal Industry											
21	PT Andalan Metal Industry											
22	PT Bintang Timur Steel											
23	PT Jiu Long Metal Industry											
24	PT Perkasa Metal Industry											

- Produksi

	A	B	C	D	E	F	G	H	I	J	K	L
1	tanggal	kode_perusah	nama_perusahaan	jenis_izin	komoditas	produk	kadar_seny	kadar_final	satuan	hasil_produksi	provinsi_produksi	pro
2	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.34	ton	32	Sulawesi Tenggara	Sult
3	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Saprolite	Ni	1.50	ton	800	Sulawesi Tenggara	Sult
4	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.44	ton	480	Sulawesi Tenggara	Sult
5	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.46	ton	5016	Sulawesi Tenggara	Sult
6	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.34	ton	1875	Sulawesi Tenggara	Sult
7	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.44	ton	703	Sulawesi Tenggara	Sult
8	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.37	ton	1979	Sulawesi Tenggara	Sult
9	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.43	ton	577	Sulawesi Tenggara	Sult
10	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.34	ton	6020	Sulawesi Tenggara	Sult
11	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Saprolite	Ni	1.59	ton	712	Sulawesi Tenggara	Sult
12	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.36	ton	4459	Sulawesi Tenggara	Sult
13	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.25	ton	1618	Sulawesi Tenggara	Sult
14	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.36	ton	8543	Sulawesi Tenggara	Sult
15	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Saprolite	Ni	1.57	ton	2306	Sulawesi Tenggara	Sult
16	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.32	ton	8197	Sulawesi Tenggara	Sult
17	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.48	ton	1246	Sulawesi Tenggara	Sult
18	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.32	ton	4511	Sulawesi Tenggara	Sult
19	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Saprolite	Ni	1.50	ton	395	Sulawesi Tenggara	Sult
20	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.29	ton	3255	Sulawesi Tenggara	Sult
21	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.19	ton	1226	Sulawesi Tenggara	Sult
22	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.33	ton	5221	Sulawesi Tenggara	Sult
23	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.44	ton	736	Sulawesi Tenggara	Sult
24	2025-01-29	63	ADHI KARTIKO PRATAMA	IUP	Nikel	Limonite	Ni	1.30	ton	6816	Sulawesi Tenggara	Sult

- Pemasaran_1

[illegible]

- Pemasaran_2

	A	B	C	D	E	F	G	H	
	tanggal	code_peruse	nama_perusahaan	jenis_perusahaan	kategori_pembeli	nama_pembeli	komoditas	produk_murni	sat
1	2026-01-21	TRADER-10003	AGI BUMI INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
2	2026-03-05	TRADER-10003	AGI BUMI INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
3	2026-01-20	TRADER-10004	ANUGERAH MINING INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
4	2026-01-20	TRADER-10004	ANUGERAH MINING INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
5	2026-02-26	TRADER-10004	ANUGERAH MINING INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
6	2026-02-27	TRADER-10004	ANUGERAH MINING INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
7	2026-03-10	TRADER-10004	ANUGERAH MINING INDONESIA	trader	smelter	PT OBSIDIAN STAINLESS STEEL	Nikel	Limonite	ton
8	2026-01-19	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
9	2026-01-26	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
10	2026-01-27	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
11	2026-01-30	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
12	2026-02-04	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
13	2026-02-14	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
14	2026-02-24	TRADER-10007	ANUGRAH NEO RESOURCES	trader	smelter	PT. INDONESIA RUPU NICKEL AND CHROME ALLOY	Nikel	Limonite	ton
15	2026-03-07	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
16	2026-03-11	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
17	2026-03-13	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
18	2026-03-17	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
19	2026-03-27	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
20	2026-04-20	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
21	2026-04-20	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
22	2026-04-20	TRADER-10008	ARTA POWERINDO SINERGY	trader	smelter	PT SHUOSHI INDONESIA INVESTMENT	Nikel	Limonite	ton
23	2026-02-11	TRADFER-10011	BARA INDAH SINERGI	trader	smelter	PT. YOUSHAN NICKEL INDONESIA	Nikel	Limonite	ton

- Perusahaan_cadangan

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	tahun	kode_perusahaan	nama_perusahaan	jenis_perusahaaa	komoditas	satuan	produk	TERKIRA	TERBUKTI	total_cadangan	total_produksi			
2	2025	63	ADHI KARTIKO PRATAMA	IUP	Nikel	ton	Limonite	3000000	1000000	1300000	0			
3	2025	63	ADHI KARTIKO PRATAMA	IUP	Nikel	ton	Saprolite	3000000	1000000	1300000	0			
4	2025	7913	VALE INDONESIA	IUPK	Nikel	ton	Nickel Matte	100000	100000	200000	0			
5	2025	8006	WANATIARA PERSADA	IUP	Nikel	ton	Ferro Nickel (FeNi)	200000	300000	500000	0			
6	2025	8025	WEDA BAY NICKEL	KK	Nikel	ton	Ferro Nickel (FeNi)	500000	500000	1000000	0			
7														
8														
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+ ≡ 🗺 ummary_2 Perizinan Data Hilirisasi Produksi Pemasaran_1 Pemasaran_2 Perusahaan_Cadangan Sumber Daya dan Cadangan Utilitas_InVENTORY < >														

- Sumber Daya dan Cadangan

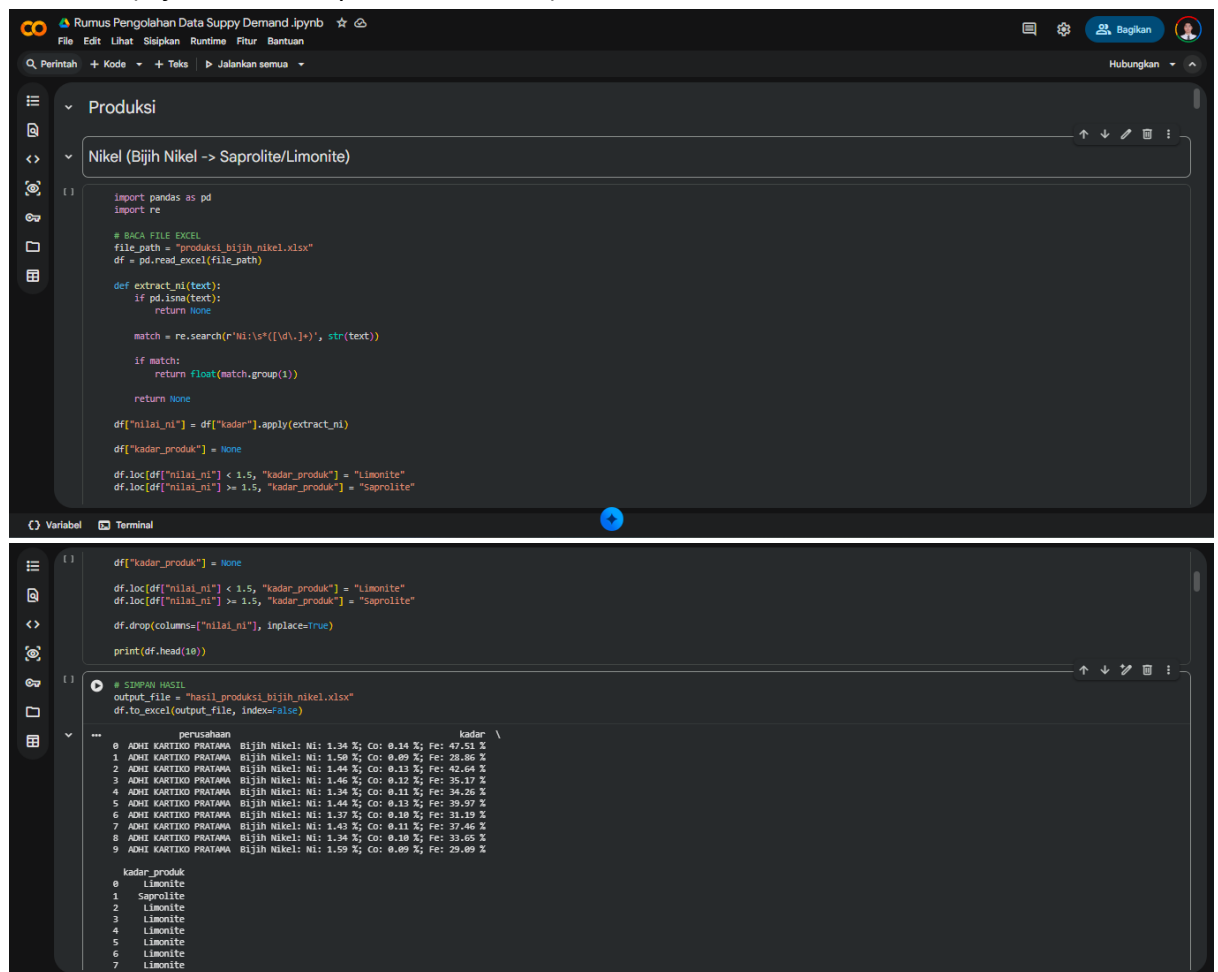
[illegible]

- Utilitas_Inventory

[illegible]

2. Google Collab

- Produksi
- Nikel (Bijih Nikel -> Saprolite/Limonite)



The screenshot displays a Google Colab environment with a Jupyter Notebook titled "Produksi". The notebook is divided into two cells. The first cell contains code for reading an Excel file, defining a function to extract nickel values from text, and applying this function to a DataFrame. The second cell contains code for classifying the data based on nickel values and saving the result to a new Excel file. The output of the second cell shows a DataFrame with columns for company name, nickel grade, and product type.

```
import pandas as pd
import re

# BACA FILE EXCEL
file_path = "produksi_bijih_nikel.xlsx"
df = pd.read_excel(file_path)

def extract_ni(text):
    if pd.isna(text):
        return None

    match = re.search(r'Ni:\s*([0-9.]+)', str(text))

    if match:
        return float(match.group(1))

    return None

df["nilai_ni"] = df["kadar"].apply(extract_ni)
df["kadar_produk"] = None

df.loc[df["nilai_ni"] < 1.5, "kadar_produk"] = "Limonite"
df.loc[df["nilai_ni"] >= 1.5, "kadar_produk"] = "Saprolite"

df["kadar_produk"] = None

df.loc[df["nilai_ni"] < 1.5, "kadar_produk"] = "Limonite"
df.loc[df["nilai_ni"] >= 1.5, "kadar_produk"] = "Saprolite"

df.drop(columns=["nilai_ni"], inplace=True)
print(df.head(10))

# SIMPAN HASIL
output_file = "hasil_produksi_bijih_nikel.xlsx"
df.to_excel(output_file, index=False)
```

	perusahaan	Bijih Nikel:	Ni:	Co:	Fe:	kadar
0	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.34 %	Co: 0.14 %	Fe: 47.51 %	
1	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.58 %	Co: 0.09 %	Fe: 28.86 %	
2	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.44 %	Co: 0.13 %	Fe: 42.64 %	
3	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.46 %	Co: 0.12 %	Fe: 35.17 %	
4	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.34 %	Co: 0.11 %	Fe: 34.26 %	
5	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.44 %	Co: 0.13 %	Fe: 39.97 %	
6	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.37 %	Co: 0.10 %	Fe: 31.19 %	
7	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.43 %	Co: 0.11 %	Fe: 37.46 %	
8	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.34 %	Co: 0.10 %	Fe: 33.65 %	
9	ADHI KARTIKO PRATAMA	Bijih Nikel:	Ni: 1.59 %	Co: 0.09 %	Fe: 29.09 %	

	kadar_produk
0	Limonite
1	Saprolite
2	Limonite
3	Limonite
4	Limonite
5	Limonite
6	Limonite
7	Limonite

- Kadar Persentase (kecuali Tembaga & Emas)



The screenshot displays a Google Colab environment with a Jupyter Notebook titled "Kadar Persentase (kecuali Tembaga & Emas)". The notebook contains a single cell with code for reading an Excel file, defining a function to extract percentage values from text, and applying this function to a DataFrame. The code uses a regular expression to match the percentage values and returns them as a pandas Series.

```
import pandas as pd
import re

# BACA FILE EXCEL
file_path = "produksi_kadar_bauksit.xlsx"
df = pd.read_excel(file_path)

target_senyawa = input(
    "Masukkan kadar senyawa : "
).strip()

def extract_senyawa(text, target):
    if pd.isna(text):
        return pd.Series([None, None, None])

    pattern = rf'{re.escape(target)}\s*([0-9.]+)\s*%([gpr])?'
    match = re.search(pattern, str(text), re.IGNORECASE)

    if match:
        nilai = match.group(1)
        satuan = match.group(2)

        return pd.Series([target, nilai, satuan])

    return pd.Series([None, None, None])

df[["kadar_senyawa", "kadar_final", "satuan"]] = df["kadar"].apply(
    lambda x: extract_senyawa(x, target_senyawa)
)
```

```
[1] df[["kadar_senyawa", "kadar_final", "satuan"]] = df["kadar"].apply(
    lambda x: extract_senyawa(x, target_senyawa)
)

print(df.head(10))

... Masukkan kadar senyawa : Al2O3

perusahaan \
0 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
1 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
2 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
3 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
4 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
5 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
6 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
7 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
8 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)
9 ANTAM TBK UBPD TAYAN (KILAYAH KALIMANTAN BARAT)

kadar kadar_senyawa \
0 Unwashed/Crude Bauksit: Al2O3: 48.78 %; SiO2: ... Al2O3
1 Washed Bauksit: Al2O3: 48.78 %; SiO2: 0.00 %; ... Al2O3
2 Washed Bauksit: Al2O3: 48.78 %; SiO2: 0.00 %; ... Al2O3
3 Washed Bauksit: Al2O3: 48.35 %; SiO2: 0.00 %; ... Al2O3
4 Unwashed/Crude Bauksit: Al2O3: 47.12 %; SiO2: ... Al2O3
5 Unwashed/Crude Bauksit: Al2O3: 51.19 %; SiO2: ... Al2O3
6 Unwashed/Crude Bauksit: Al2O3: 51.19 %; SiO2: ... Al2O3
7 Unwashed/Crude Bauksit: Al2O3: 42.46 %; SiO2: ... Al2O3
8 Washed Bauksit: Al2O3: 50.10 %; SiO2: 0.00 %; ... Al2O3
9 Unwashed/Crude Bauksit: Al2O3: 50.38 %; SiO2: ... Al2O3

kadar_final satuan
0 48.78 %
1 48.78 %
2 48.78 %
3 48.35 %
4 47.12 %
5 51.19 %
6 51.19 %
7 42.46 %
8 50.10 %
9 50.38 %

[1] # SIMPAN HASIL
output_file = f"hasil_kadar_{target_senyawa}.xlsx"
df.to_excel(output_file, index=False)
```

- Tembaga - Kadar Presentase - Ton/toz

```
Tembağa - Kadar Presentase - Ton/toz

[1] import pandas as pd
import re

# BACA FILE
file_path = "produksi_mantahan_tembaga.xlsx"
df = pd.read_excel(file_path)

def get_kadar_final(kadar_text, target):
    if pd.isna(kadar_text):
        return pd.Series([None, None])

    pattern = rf'^(target){s}*{s}([d\.,]+){s}*{s}([gr/t])$'
    match = re.search(pattern, str(kadar_text), re.IGNORECASE)

    if match:
        nilai = match.group(1)
        satuan = match.group(2)
        return pd.Series([nilai, satuan])

    return pd.Series([None, None])

def split_data(row):
    hasil = []

    ekuivalen_text = str(row["ekuivalen"])
    items = ekuivalen_text.split("|")
```

```
f1 def split_data(row):
    hasil = []

    ekuivalen_text = str(row["ekuivalen"])
    items = ekuivalen_text.split("|")

    for item in items:
        item = item.strip()

        pattern = r"^(.+?)\\s*:\\s*(Cu|Au|Ag|Te|Pt|P|Ru|Ir|Rh)\\s*:\\s*{([\\d\\.]+)}\\s*(Ton|toz)"
        matches = re.finditer(pattern, item, re.IGNORECASE)

        for match in matches:
            new_row = row.copy()

            produk = match.group(1).strip()
            kadar = match.group(2).strip()
            ekuivalen = match.group(3).strip()
            satuan_ekuivalen = match.group(4).strip()

            kadar_final, satuan_kadar = get_kadar_final(
                row["kadar_final"],
                kadar
            )

            new_row["produk"] = produk
            new_row["kadar"] = kadar
            new_row["ekuivalen"] = ekuivalen
            new_row["satuan"] = satuan_ekuivalen
            new_row["kadar_final"] = kadar_final
            new_row["satuan_kadar"] = satuan_kadar

            hasil.append(new_row)
```

```
[1]
hasil.append(new_row)

return hasil

rows_baru = []

for _, row in df.iterrows():
    hasil_split = split_data(row)

    if hasil_split:
        rows_baru.extend(hasil_split)
    else:
        rows_baru.append(row)

df_final = pd.DataFrame(rows_baru)

print(df_final.head(10))
```

```
...
   tanggal_laporan  perusahaan  produk
0  2025-12-09 14:56:32  AMMAN MINERAL NUSA TENGGARA  Bijih Tembaga
1  2025-12-09 15:55:05  AMMAN MINERAL NUSA TENGGARA  Katoda Tembaga
2  2025-12-09 15:55:05  AMMAN MINERAL NUSA TENGGARA  Lumpur Anoda
2  2025-12-10 11:33:19  AMMAN MINERAL NUSA TENGGARA  Konsentrat Tembaga
3  2025-12-10 11:45:49  AMMAN MINERAL NUSA TENGGARA  Emas (Asosiasi Tembaga)
3  2025-12-10 11:45:49  AMMAN MINERAL NUSA TENGGARA  Perak (Asosiasi Tembaga)
4  2025-12-10 11:39:09  AMMAN MINERAL NUSA TENGGARA  Bijih Tembaga
5  2025-12-10 11:36:31  AMMAN MINERAL NUSA TENGGARA  Katoda Tembaga
5  2025-12-10 11:36:31  AMMAN MINERAL NUSA TENGGARA  Lumpur Anoda
6  2025-12-10 11:54:17  AMMAN MINERAL NUSA TENGGARA  Konsentrat Tembaga

satuan  provinsi  ekuivalen  kadar_final  kadar  satuan_kadar
0  Ton  Nusa Tenggara Barat  18029.728636  0.37  Cu  %
1  Ton  Nusa Tenggara Barat  2302.661610  0.00  Fe  %
```

```
5 2025-12-10 11:36:31 AMMAN MINERAL MUSA TENGGARA Lumpur Anoda
6 2025-12-10 11:54:17 AMMAN MINERAL MUSA TENGGARA Konsentrat Tembaga

satsun provinsi ekuivalen kadar_final kadar satuan_kadar
0 Ton Nusa Tenggara Barat 18028.72086 0.37 Cu %
1 Ton Nusa Tenggara Barat 3258.404159 99.99 Au %
1 Ton Nusa Tenggara Barat 0.232320 0.88 Au %
2 Ton Nusa Tenggara Barat 2565.759986 21.62 Cu %
3 ton Nusa Tenggara Barat 6879.197020 99.99 Au %
3 ton Nusa Tenggara Barat 16743.224200 99.95 Ag %
4 Ton Nusa Tenggara Barat 9468.130534 0.37 Cu %
5 Nusa Tenggara Barat 2380.424049 99.99 Cu %
5 Ton Nusa Tenggara Barat 0.283511 0.88 Au %
6 Ton Nusa Tenggara Barat 2245.040258 21.62 Cu %

# SIMPAN HASIL
output_file = "hasil_kadar_ekuivalen_tembaga.xlsx"
df_final.to_excel(output_file, index=False)
```

- Emas - Kadar Ton/toz - Inputan Auto Filter

```
Emas - Kadar Ton/toz - Inputan Auto Filter

import pandas as pd
import re

# BACA FILE
file_path = "produksi_mentahan_emas.xlsx"
df = pd.read_excel(file_path)

def get_kadar_info(text, target):

    if pd.isna(text):
        return pd.Series([None, None, None])

    pattern = rf'{target}\s*:\s*([d.]+)-\s*\s*([a-zA-ZIV]+)'+
    match = re.search(pattern, str(text), re.IGNORECASE)

    if match:

        nilai_text = match.group(1).replace(".", "")

        nilai = float(nilai_text)

        satuan = match.group(2)

        if satuan.lower() == "gr/t":
            konversi = nilai / 10000
        else:
            konversi = nilai

        return pd.Series([nilai, satuan, konversi])
```

```

return pd.Series([nilai, satuan, konversi])

return pd.Series([None, None, None])

def process_row(row):
    hasil = []
    teks = str(row["ekuivalen"])
    items = teks.split("|") if "|" in teks else [teks]
    for item in items:
        item = item.strip()

        pattern = r'^(.+?)\s*:\s*(Au|Ag)\s*\s*(\s*(\d\.\d+)?\s*)\s*(Ton|kg|toz)'
        matches = re.finditer(pattern, item, re.IGNORECASE)

        for match in matches:
            new_row = row.copy()

            produk = match.group(1).strip()
            senyawa = match.group(2).strip()

            ekuivalen = match.group(3).replace(",", ".")
            satuan_ekuivalen = match.group(4).strip()

            kadar_final, satuan_kadar, kadar_konversi = get_kadar_info(
                row["kadar"],
                senyawa

```

```

kadar_final, satuan_kadar, kadar_konversi = get_kadar_info(
    row["kadar"],
    senyawa
)

new_row["produk"] = produk
new_row["kadar_senyawa"] = senyawa

new_row["ekuivalen"] = ekuivalen
new_row["satuan_ekuivalen"] = satuan_ekuivalen

new_row["kadar_final"] = kadar_final
new_row["satuan_kadar"] = satuan_kadar
new_row["kadar_konversi"] = kadar_konversi

hasil.append(new_row)

if not hasil:
    hasil.append(row)

return hasil

rows_baru = []
for _, row in df.iterrows():
    rows_baru.extend(process_row(row))

df_final = pd.DataFrame(rows_baru)

df_final["kadar_konversi"] = (
    df_final["kadar_konversi"]
    .astype(str)
    .str.replace(",", ".", regex=False)

```

```

df_final["kadar_konversi"] = (
    df_final["kadar_konversi"]
    .astype(str)
    .str.replace(",", ".", regex=False)
)

kolom_urut = [
    "tanggal_laporan",
    "perusahaan",
    "produk",
    "hasil_produksi",
    "provinsi",
    "kadar_senyawa",
    "ekuivalen",
    "satuan_ekuivalen",
    "kadar_final",
    "satuan_kadar",
    "kadar_konversi"
]

kolom_urut = [col for col in kolom_urut if col in df_final.columns]

df_final = df_final[kolom_urut]

print(df_final.head())

```

	tanggal_laporan	perusahaan	produk	hasil_produksi
0	2025-12-30 14:59:00	AGINCOURT RESOURCES	Bijih Emas	448400
1	2025-12-30 14:59:00	AGINCOURT RESOURCES	Bijih Emas	0
2	2025-12-30 15:00:00	AGINCOURT RESOURCES	Bijih Emas	0
3	2025-12-30 15:15:00	AGINCOURT RESOURCES	Bijih Emas	310300
4	2025-12-30 15:30:00	AGINCOURT RESOURCES	Bijih Emas	0

```

provinsi kadar_senyawa ekuivalen satuan_ekuivalen kadar_final \
0 Sumatera Utara Au 0.000009 Ton 1.34
1 Sumatera Utara Au 0.000000 Ton 0.00
2 Sumatera Utara Au 0.000000 Ton 0.00
3 Sumatera Utara Au 0.002110 Ton 0.68
4 Sumatera Utara Au 0.000000 Ton 0.00

satuan_kadar kadar_konversi
0 gr/T 0.000134
1 gr/T 0.0
2 gr/T 0.0
3 gr/T 6.3e-05
4 gr/T 0.0

# SIMPAN
output_file = "hasil_split_emas.xlsx"
df_final.to_excel(output_file, index=False)

```

- Pemasaran
- Hapus Nilai Duplikat volume_final

```

Pemasaran
Hapus Nilai Duplikat volume_final

import pandas as pd

df = pd.read_excel("pemasaran_volume_final_nikel.xlsx")

df["volume_final"] = (
    df["volume_final"]
    .astype(str)
    .str.replace(",", ".", regex=False)
)

df["volume_final"] = pd.to_numeric(
    df["volume_final"],
    errors="coerce"
)

volume_ntpn = {}

for ntpn, grp in df.groupby("ntpn"):
    iup_rows = grp[
        grp["jenis_perusahaan"]
        .astype(str)
        .str.upper()
        .eq("IUP")
    ]

```

```

for ntpn, grp in df.groupby("ntpn"):
    iup_rows = grp[
        grp["jenis_perusahaan"]
        .astype(str)
        .str.upper()
        .eq("IUP")
    ]

    if not iup_rows.empty:
        volume_ntpn[ntpn] = iup_rows.iloc[0]["volume_final"]
    else:
        volume_ntpn[ntpn] = grp.iloc[0]["volume_final"]

df["volume_final_clean"] = df["volume_final"]

for _, grp in df.groupby("ntpn"):
    iup_rows = grp[
        grp["jenis_perusahaan"]
        .astype(str)
        .str.upper()
        .eq("IUP")
    ]

    if len(iup_rows) > 0:
        keep_idx = iup_rows.index[0]
        zero_idx = grp.index.difference([keep_idx])

        df.loc[zero_idx, "volume_final_clean"] = 0

df["trader_ke"] = pd.NA

```

```

df["trader_ke"] = pd.NA

priority_main = ["IUP", "IUPK", "IK"]

for ntpn, grp in df.groupby("ntpn"):
    trader_rows = grp[
        grp["kategori_pembeli"]
        .astype(str)
        .str.lower()
        .eq("trader")
    ].copy()

    if trader_rows.empty:
        continue

    trader_rows["prioritas"] = trader_rows["jenis_perusahaan"].apply(
        lambda x: 0 if str(x).upper() in priority_main else 1
    )

    trader_rows = trader_rows.sort_values(
        ["prioritas"],
        kind="stable"
    )

    for i, idx in enumerate(trader_rows.index, start=1):
        df.loc[idx, "trader_ke"] = i

max_trader = int(df["trader_ke"].dropna().max()) if df["trader_ke"].notna().any() else 0

for i in range(1, max_trader + 1):

```



```

for i in range(1, max_trader + 1):
    col = f"volume_final_trader_{i}"
    df[col] = 0

    mask = df["trader_ke"] == i

    df.loc[mask, col] = (
        df.loc[mask, "ntpn"].map(volume_ntpn)
    )

    df["volume_final_smelter"] = 0

    mask_smelter = (
        df["kategori_pembeli"]
        .astype(str)
        .str.lower()
        .eq("smelter")
    )

    df.loc[mask_smelter, "volume_final_smelter"] = (
        df.loc[mask_smelter, "ntpn"].map(volume_ntpn)
    )

df.to_excel("hasil_pemesaran_volume_final_nikel.xlsx", index=False)

```

- Nikel (Kadar -> Saprolite/Limonite)

```

import pandas as pd
import re

file_path = "pemesaran_nikel.xlsx"
df = pd.read_excel(file_path)

df["kadar_final"] = df["kadar"].str.extract(
    r'(Ni:\s*\d+(?:\.\d+)?\s*\%)',
    expand=False
)

def extract_ni_value(text):
    if pd.isna(text):
        return None

    match = re.search(r'Ni:\s*(\d+(?:\.\d+)?)', str(text))

    if match:
        return float(match.group(1))

    return None

df["nilai_ni"] = df["kadar"].apply(extract_ni_value)

df["kadar_produk"] = None

df.loc[df["nilai_ni"] < 1.5, "kadar_produk"] = "Limonite"
df.loc[df["nilai_ni"] >= 1.5, "kadar_produk"] = "Saprolite"

```

```

df.loc[df["nilai_ni"] < 1.5, "kadar_produk"] = "Limonite"
df.loc[df["nilai_ni"] >= 1.5, "kadar_produk"] = "Saprolite"

df.drop(columns=["nilai_ni"], inplace=True)

print(df.head(10))

...
perusahaan      kadar kadar_final \
0 ADHI KARTIKO PRATAMA  Ni: 1.25 %; Co: 0.13 %; Fe: 38.87 %  Ni: 1.25 %
1 ADHI KARTIKO PRATAMA  Ni: 1.53 %; Co: 0.06 %; Fe: 22.58 %  Ni: 1.53 %
2 ADHI KARTIKO PRATAMA  Ni: 1.33 %; Co: 0.10 %; Fe: 35.89 %  Ni: 1.33 %
3 ADHI KARTIKO PRATAMA  Ni: 1.54 %; Co: 0.07 %; Fe: 26.02 %  Ni: 1.54 %
4 ADHI KARTIKO PRATAMA  Ni: 1.23 %; Co: 0.11 %; Fe: 37.18 %  Ni: 1.23 %
5 ADHI KARTIKO PRATAMA  Ni: 1.31 %; Co: 0.10 %; Fe: 36.15 %  Ni: 1.31 %
6 ADHI KARTIKO PRATAMA  Ni: 1.32 %; Co: 0.11 %; Fe: 37.39 %  Ni: 1.32 %
7 ADHI KARTIKO PRATAMA  Ni: 1.29 %; Co: 0.11 %; Fe: 36.33 %  Ni: 1.29 %
8 ADHI KARTIKO PRATAMA  Ni: 1.24 %; Co: 0.13 %; Fe: 35.66 %  Ni: 1.24 %
9 ADHI KARTIKO PRATAMA  Ni: 1.60 %; Co: 0.05 %; Fe: 23.24 %  Ni: 1.60 %

kadar_produk
0 Limonite
1 Saprolite
2 Limonite
3 Saprolite
4 Limonite
5 Limonite
6 Limonite
7 Limonite
8 Limonite
9 Saprolite

df.to_excel("hasil_pemesaran_nikel.xlsx", index=False)

```

- Emas - Ekuivalen Final

```

Emas - Ekuivalen Final

import pandas as pd
import re

file_path = "emas_ekuivalen_final.xlsx"
df = pd.read_excel(file_path)

def extract_ekuivalen_final(text):
    if pd.isna(text):
        return pd.Series([None, None, None])

    pattern = r'([A-Za-z0-9]+)\s*:\s*([d\.,]+)\s*([A-Za-z\./]+)'
    match = re.search(pattern, str(text), re.IGNORECASE)

    if match:
        kadar = match.group(1).strip()
        volume_lhv = match.group(2).strip()
        satuan = match.group(3).strip()

        return pd.Series([kadar, volume_lhv, satuan])

    return pd.Series([None, None, None])

df[["kadar", "volume_lhv", "satuan"]] = df["ekuivalen_final"].apply(
    extract_ekuivalen_final
)

print(df.head())

```

```
print(df.head())
```

```
df.to_excel("hasil_ekuivalen_final.xlsx", index=False)
```

	ekuivalen_final	kadar	volume_lhv	satuan
0	Au: 5888.971852 toz	Au	5888.971852	toz
1	Au: 3921.667794 toz	Au	3921.667794	toz
2	Ag: 3234.044890 toz	Ag	3234.044890	toz
3	Au: 329.517045 toz	Au	329.517045	toz
4	Au: 313.438653 toz	Au	313.438653	toz

- Tembaga - Ekuivalen Final

```

Tembaga - Ekuivalen Final

import pandas as pd
import re

file_path = "tembaga_ekuivalen_final.xlsx"
df = pd.read_excel(file_path)

def extract_ekuivalen_final(text):
    if pd.isna(text):
        return pd.Series([None, None, None])

    text = str(text).strip()

    # Ambil pasangan pertama saja
    pattern = r'^([A-Za-z0-9]+)\s*:\s*([d\.,]+)\s*([A-Za-z\./]+)'
    match = re.search(pattern, text)

    if match:
        kadar = match.group(1).strip()
        volume_lhv = match.group(2).strip()
        satuan = match.group(3).strip()

        return pd.Series([kadar, volume_lhv, satuan])

    return pd.Series([None, None, None])

```

```
return pd.Series([None, None, None])
```

```
df[["kadar", "volume_lhv", "satuan"]] = df["ekuivalen_final"].apply(
    extract_ekuivalen_final
)

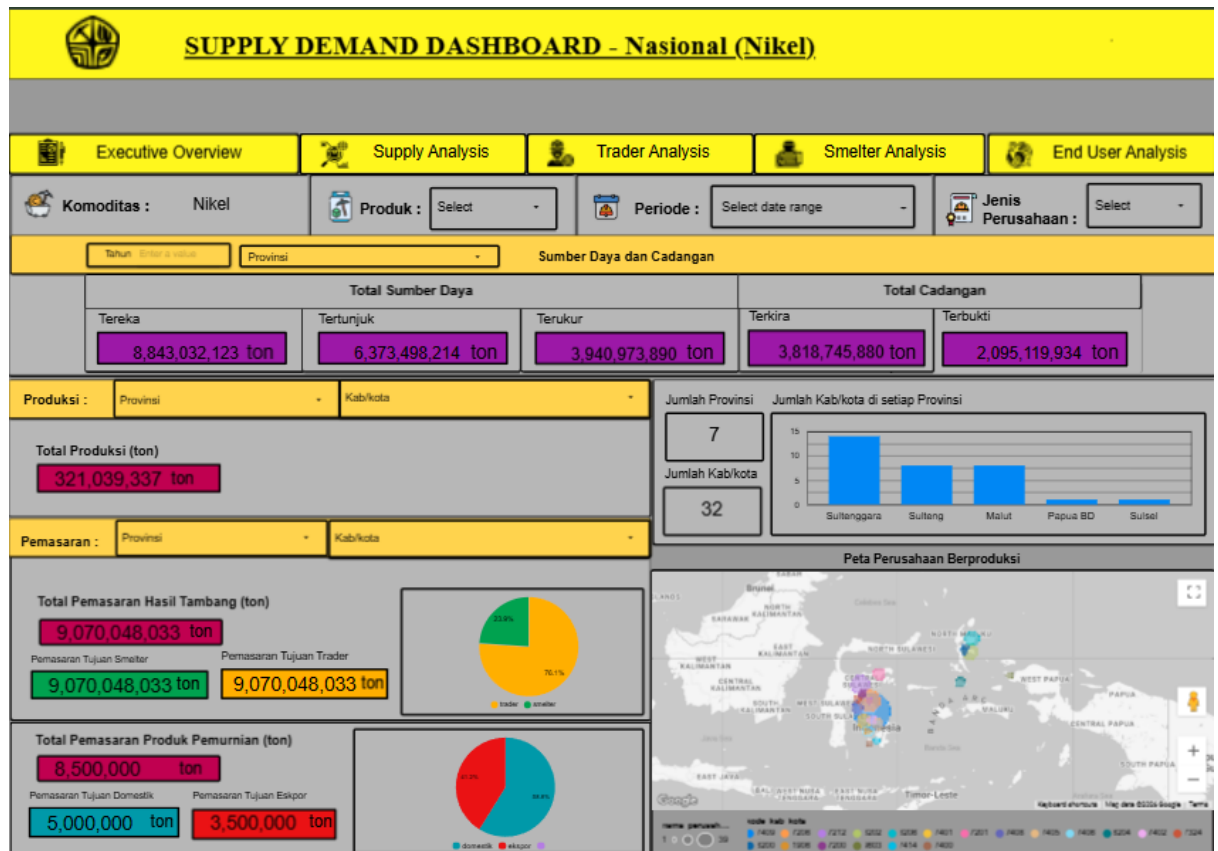
print(df.head())
```

```
df.to_excel("hasil_ekuivalen_final2.xlsx", index=False)
```

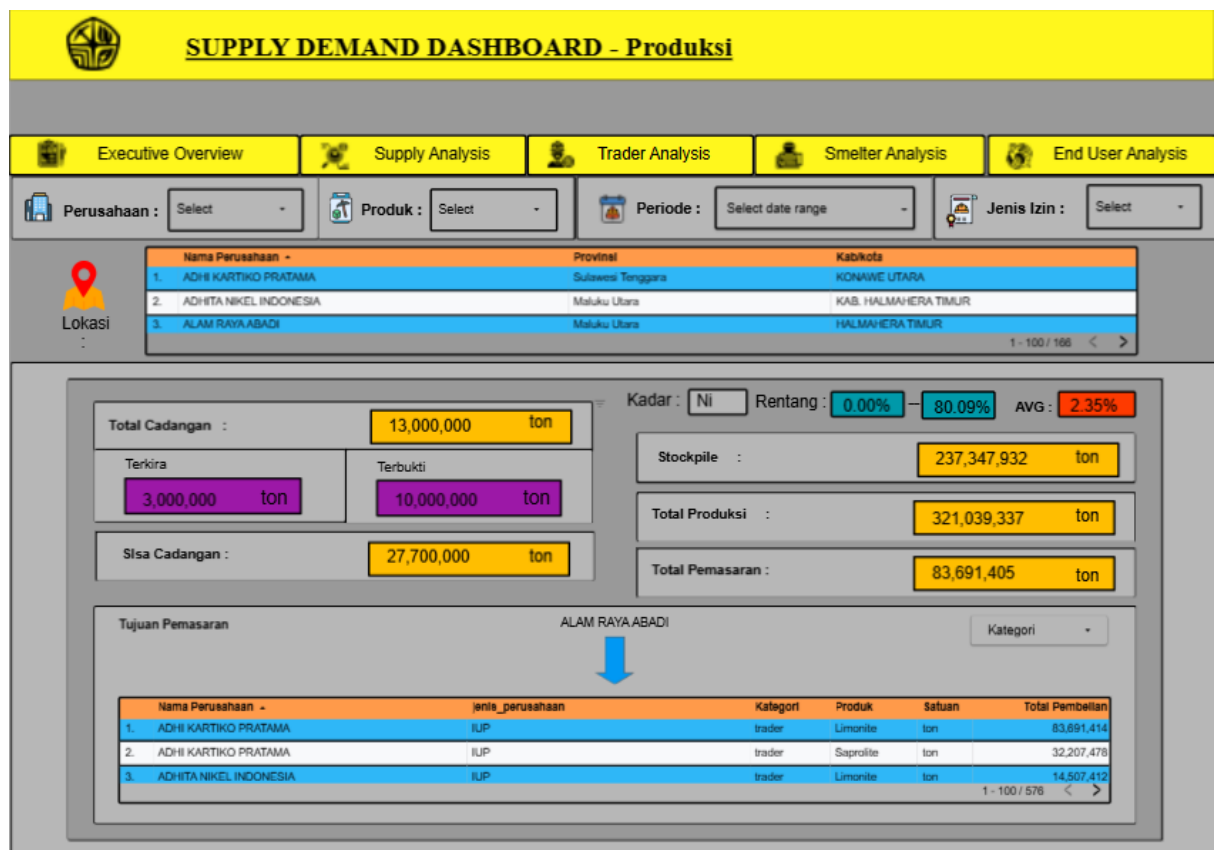
	ekuivalen_final	kadar	volume_lhv	satuan
0	Cu: 296.700327 Ton	Cu	296.700327	Ton
1	Cu: 196.730325 Ton	Cu	196.730325	Ton
2	Cu: 199.859013 Ton	Cu	199.859013	Ton
3	Cu: 147.585240 Ton	Cu	147.585240	Ton
4	Cu: 197.348264 Ton	Cu	197.348264	Ton

3. Looker Studio

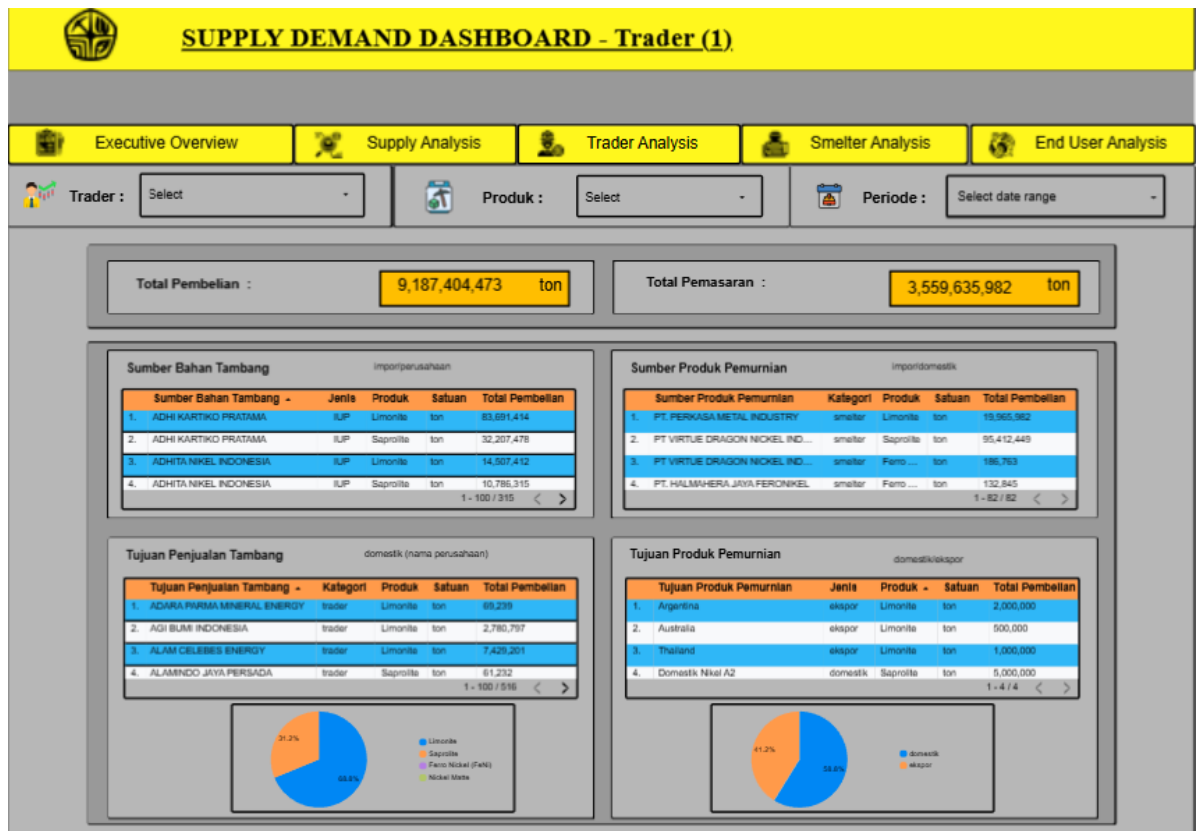
- Nasional



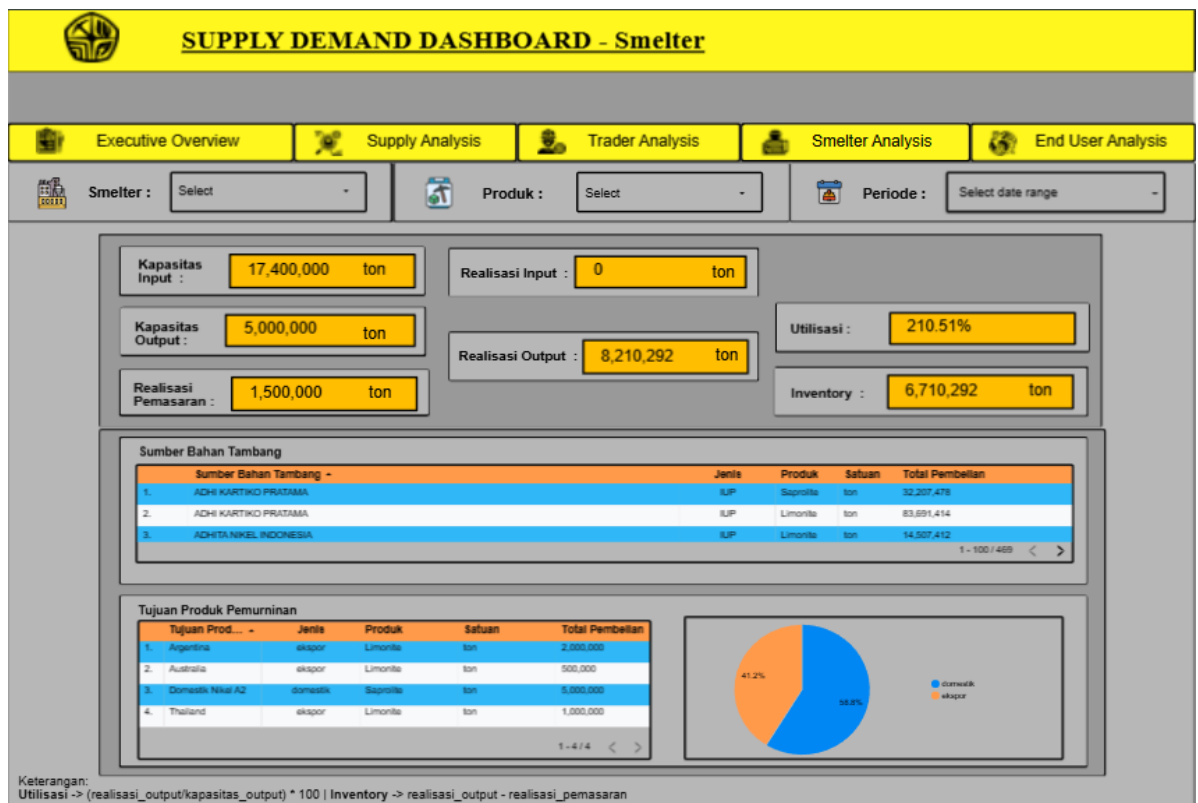
- Produksi



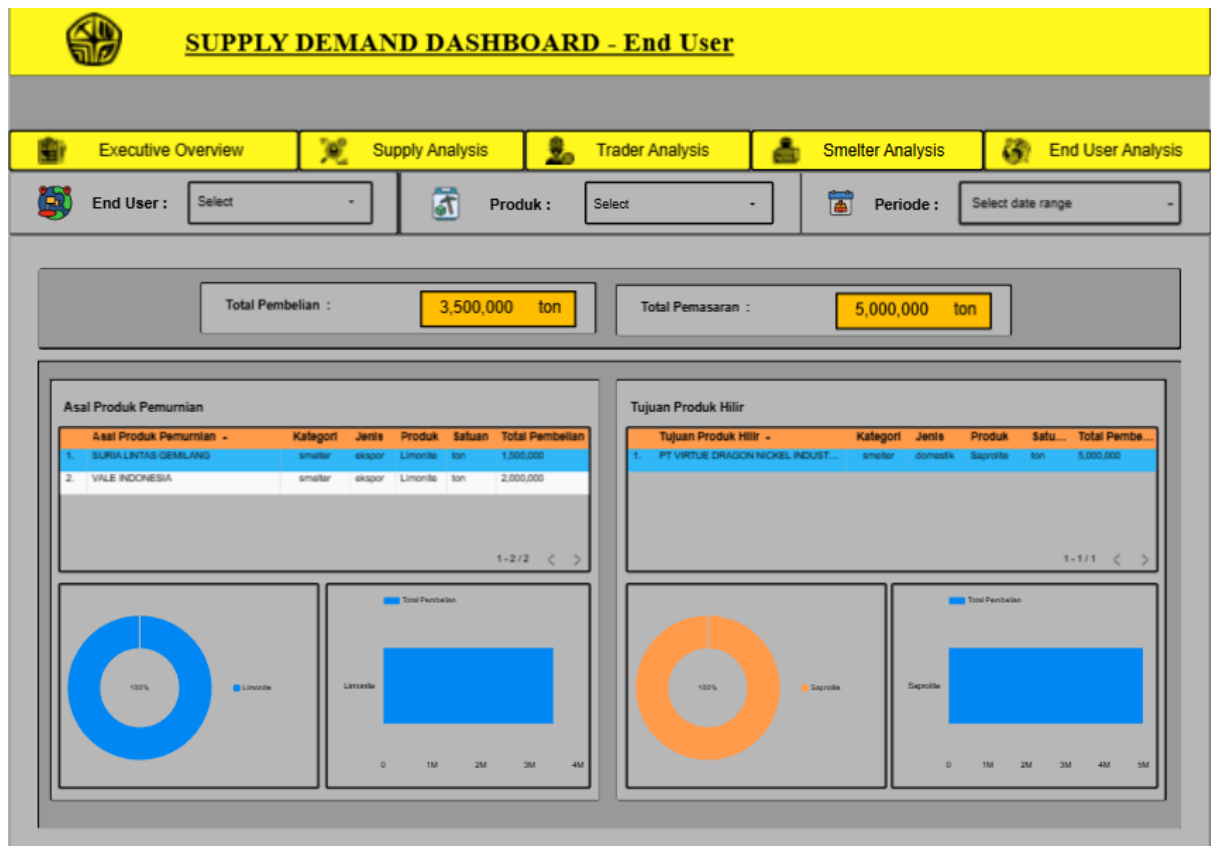
- Pemasaran
 - ➔ Trader



- ➔ Smelter



➔ End User



4. Mindmap

